

Social displays

Social displays take many different forms, from threat display to courtship and bonding. Both cuttlefish and octopuses can change colour; they darken and flash different colours to intimidate rivals or enemies. The male Uganda kob, a type of antelope, establishes territorial breeding grounds by displaying along the boundary of his territory. Lowering his head, he makes a mock attack with his horns. This warns rival males to keep out of his territory, while at the same time, induces other females to join his harem.



Tail feathers overlap and rest on the ground when relaxed.

Peacock

Courtship

Many animals use courtship displays to attract a mate. The fiddler crab, for example, waves its outside claw, the elephant seal inflates its nose, and the grouse spreads its tail and inflates its air sacs. Among the most impressive courtship displays are that of the male peacock, which spreads his brilliantly coloured tail plumage, and the elaborate rituals of birds-of-paradise and bowerbirds. These involve vibrating the body, fanning feathers, puffing out plumage, decorating nesting areas, and calling loudly.



Peacock starting to erect tail plumage.

Male calls as he starts to display.



Ring-tailed lemurs signalling with raised tails

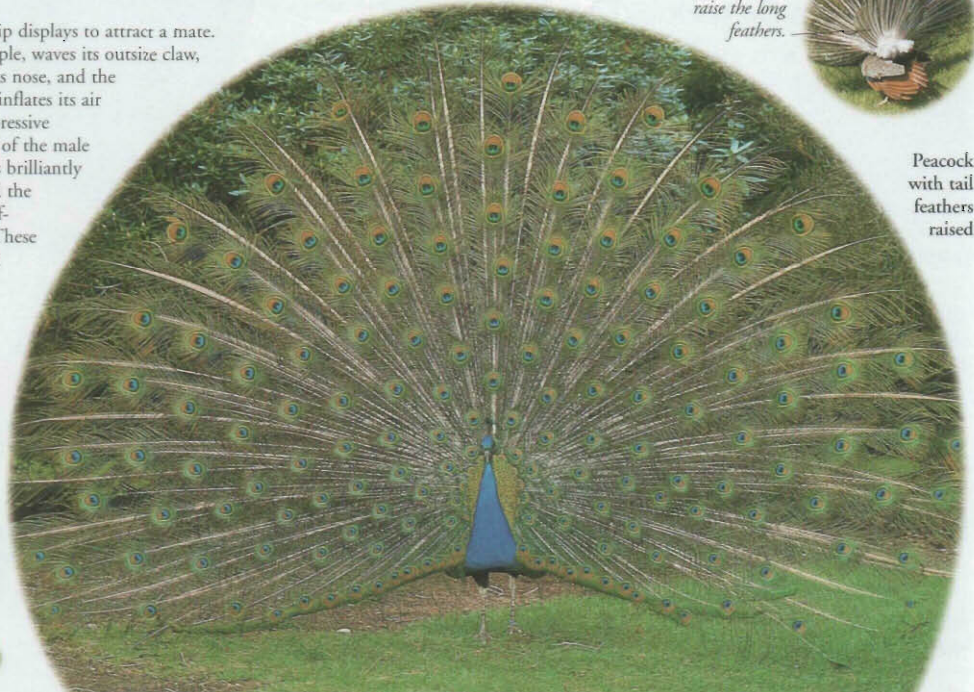
Signalling

Signs and signals help animals to maintain contact, preserve the social hierarchy, and intimidate rivals and enemies. The signals have to be conspicuous and unmistakable. The ring-tailed lemurs of Madagascar raise their long black-and-white tails to waft scent at their rivals, and to enable all members of a group to maintain contact. The black rings encircling the cheetah's white-tipped tail enable the cubs to follow their parent, which would otherwise be invisible in the long grass. The young of ringed plovers have a white neck-band which helps the parents keep the brood together.

Strong feathers at the rear, attached to muscles, are used to raise the long feathers.

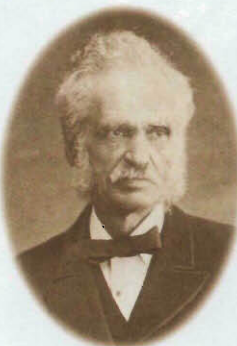


Peacock with tail feathers raised



Henry Walter Bates

The English naturalist and explorer, Henry Bates (1825–92) spent 11 years exploring the Amazon, returning with 8,000 species of previously unknown insects. In 1861, he published a paper on mimicry which made an important contribution to the theory of natural selection. He suggested that some harmless insects looked like harmful ones to discourage predators from attacking them.



Red and black frog hopper

False warning

Many animals employ bluff as a means of defence. In birds, this may take the form of fluffing up feathers, spreading wings, and clacking beaks. Many frogs and toads blow themselves up, to make them appear larger; the hawkmoth caterpillar looks like a snake to intimidate enemies; and the Australian frilled lizard erects its frill and hisses loudly to intimidate intruders.

Warning signals

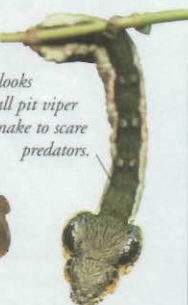
Animals use many methods to frighten off other animals. Warning colours make prey appear unpalatable to discourage predators. Many poisonous and venomous animals do not need to be camouflaged; they advertise themselves with bright coloured patterns of red, yellow, and black, which are recognized warning colours. Skunks' black and white coats warn they can squirt foul-smelling spray.



The toad raises itself on its legs to make itself appear bigger.

European common toad

The caterpillar looks like a small pit viper snake to scare predators.



Hawkmoth caterpillar

Seasonal change

Some Arctic animals, such as the polar bear and snowy owl, remain white throughout the year; others undergo a seasonal change. In far-northern latitudes, the stoat becomes completely white in winter, except for the tip of its tail, which remains black. In the warmer parts of its habitat, it can retain its russet coloration, become part-coloured, or change to white as needed. This ability to change colour provides the stoat with effective camouflage throughout the year.



Stoat with dark summer coat



Stoat with pale winter coat

FIND OUT
MORE

BIRDS

BUGS

DEER AND
ANTELOPES

FROGS AND
TOADS

LIONS AND OTHER
WILD CATS

MONKEYS AND
OTHER PRIMATES

OWLS AND
NIGHTJARS

POISONOUS
ANIMALS

SNAKES